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Energy Balance

$$\frac{\partial}{\partial t} \sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \quad (1)$$

Local Force Balance

$$\dot{p} = F \hat{=} \underbrace{\varrho \mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force}} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}} = 0 \quad (2)$$

Local Momentum Conservation

linear momentum conservation if $p = \text{const.}$, otherwise “tension” occurs:

$$p = \int F = \int \frac{d}{dt} p \Rightarrow F = \dot{p} \hat{=} \int_{\mathbb{R}^3} \frac{d}{dt} \left[(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S} \right] = \frac{d}{dt} \underbrace{\left[\int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S} \right]}_{= p \text{ momentum conserved if } p = \text{const.}} = \begin{cases} 0 \Leftrightarrow p = \text{const.} \\ - \int_{\mathbb{R}^2} \mathcal{T} d\mathbf{x} \end{cases} \quad (3)$$

$$\text{Momentum } p = \int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$$

$$\text{photon Spin} = \int_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=} \dots \quad (4)$$

Tension Tensor

$$\nabla \cdot \mathcal{T} = \varepsilon_0 (\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E} \nabla \cdot \mathcal{E}) + \frac{1}{\mu_0} (\mathcal{B} \nabla \times \mathcal{B} - \mathcal{B} \nabla \cdot \mathcal{B}) \quad (5)$$

Abbreviations:

w_i = energy densities (electric w_e , magnetic w_m , gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectors $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Tension Tensor (momentum flow density); Tension $\hat{=}$ Forces = Pressure - Traction $\hat{=}$ Torsion Tension at angular momentum

Mechanical

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Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (6)$$

$\varrho = \sigma \delta$ = charge density in interior

σ = charge density at border (skin)

δ = Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\bar{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (7)$$

Abbreviations:

$$\Delta \cdot := \operatorname{div} \operatorname{grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogeneity, cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potential difference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \quad C = \text{Capacitance} \quad Q = \text{charge (ion)} \quad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector, radiation intensity}$$

$$D_S = \int (\cdot \mathcal{S}) \, dr = \int \operatorname{div}(\mathcal{E} \times \mathcal{H}) \, dr = \text{dissipation energy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}i\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor (dielectricum)}$$

Thermal

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Gravitational

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